

Advanced Machine Learning Application for Real Estate Price Prediction

1. Executive Summary

This project outlines the development of an end-to-end Machine Learning web application designed to predict real estate prices. By integrating multiple regression models—including Scikit-Learn (Random Forest, Gradient Boosting, Ridge Regression), XGBoost, and LightGBM—the system evaluates house features to generate accurate, real-time property valuations. The solution unifies data processing, model training, and a user-friendly frontend interface into a streamlined architecture using Streamlit.

2. Dataset Description & Preprocessing

The application utilizes a real estate dataset containing historical housing metrics. The initial dataset contained 4,600 rows and 18 columns. During data cleaning, records containing a target price of \$0 were identified and removed, yielding a clean dataset of 4,551 active records.

- **Target Variable:** price (Continuous)
- **Selected Features:** bedrooms, bathrooms, sqft_living, sqft_lot, floors, yr_built, condition, waterfront, and view.

Feature	Average / Mean	Minimum	Maximum
Price	\$557,905	\$7,800	\$26,590,000
Bedrooms	3.39	0	9
Bathrooms	2.15	0	8
Living Area (Sqft)	2,132	370	13,540
Year Built	1970	1900	2014

3. System Architecture & Methodology

The system follows a modern ML pipeline designed for robust and scalable predictions.

- **Data Splitting:** The data is split using an 80/20 ratio for training and testing to prevent overfitting.
- **Feature Scaling:** Continuous features are normalized using StandardScaler to ensure optimal performance, particularly for linear algorithms like Ridge Regression.
- **Pipeline Integration:** Scikit-Learn's Pipeline API is used to bind the scaler and the machine learning model together, ensuring that user inputs undergo the exact same transformation during inference.

4. Application Implementation

The full stack of the application is encapsulated in a unified Python environment.

- **Backend Inference:** Driven by Scikit-Learn, LightGBM, and XGBoost. The models are instantiated, trained in the background, and cached in memory to minimize latency on page reloads.
- **Frontend UI:** Built with Streamlit. The interface includes interactive sliders, drop-down menus, and number inputs grouped logically into columns.
- **Dependency Management:** All libraries and exact version parameters are bundled into a `requirements.txt` file, ensuring perfect environment replication and seamless deployment.

5. Model Evaluation & Metrics

To benchmark the system, we evaluated three primary algorithms on the testing data (20% holdout). Based on the selected feature space, the baseline evaluation yielded the following results:

Model Algorithm	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R-Squared Score
Ridge Regression	\$159,396	\$242,567	0.6045
Random Forest	\$157,354	\$254,249	0.5655
Gradient Boosting	\$162,424	\$267,177	0.5202

Analytical Note: The presence of extreme outliers in the dataset (such as properties valued at up to \$26.5M) significantly inflates the RMSE penalty across all models. Ridge Regression handled the extreme variance most robustly in the linear space, while Random Forest achieved the best absolute margin of error.

6. Conclusion & Future Scope

The current implementation successfully bridges complex machine learning infrastructure with an accessible, interactive user interface. Real-time predictions execute in milliseconds, allowing end-users to dynamically test housing scenarios.

Future Enhancements:

- **Geospatial Processing:** Incorporating city and zip code features via One-Hot Encoding to capture neighborhood pricing premiums.
- **Outlier Handling:** Implementing robust outlier capping (e.g., removing the top 1% of extreme luxury prices) to significantly boost the R-Squared correlation.
- **REST API Decoupling:** Migrating the prediction endpoint to a standalone FastAPI server for broader integration into mobile apps or external web platforms.